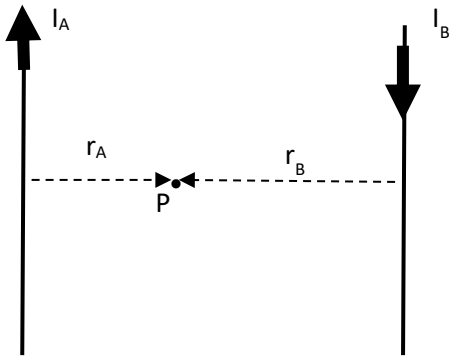
Physics 2 Problem Solving – Chapter 47 Magnetic fields and Force

1) A current carrying wires A has a current of I_A = 10 A in the upward direction and the current in wire B I_B = 12 A in the downward direction. Point P is r_A = 2 mm from wire A and r_B = 3 mm from wire B.

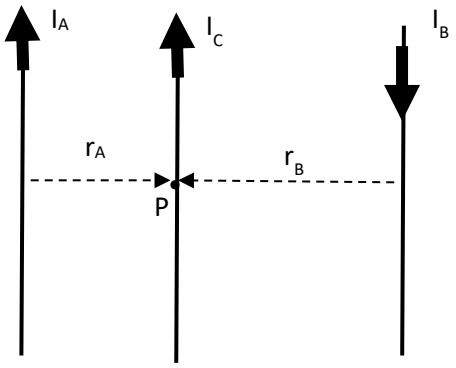


A) Determine magnitude and direction of the net magnetic field at point P.

Answer: 1.93 x 10^{-3} T directed into the page (-$\hat{z}$)

B) A third wire C is placed parallel to wire A and B at the position of point P with a current of IC = 12 A upward and is L = 0.5 m long. Determine the magnitude and direction of the magnetic force exerted on wire C due to wire A and B.

Answer: 0.012 N to the left



2) Two current carrying circular loops (N = 1) are placed one inside the other, Loop A is the outside loop and has a radius of 0.1 m and a counterclockwise current of I_A = 10 A. Loop B is inside loop A and has a radius of R_B = 0.05 m. Determine the magnitude and direction of the current in loop B in order for the net magnetic field to equal zero at the center of the loops.

Answer: 5 A in the clockwise direction.

